By Lemma 2.44, choose $a_0 \in D$ at which $F$ is continuous, and put $E = F(a_0)$. After shrinking $D$ around $a_0$, continuity gives

$$F(a) \subseteq \bigcup_{e \in E} \mathfrak{o}(e, \gamma) \qquad (a \in D).$$

Here $E$ is finite and nonempty. The finitely many definable sets

$$D_e = \{a \in D : F(a) \cap \mathfrak{o}(e, \gamma) \neq \varnothing\}, \qquad e \in E,$$

cover $D$. Hence one of them contains an open subdisc $D'$. Fix its index $e$. For each $a \in D'$, choose $x \in F(a) \cap \mathfrak{o}(e, \gamma)$. Since $(a, x) \in T_\gamma$ and intersecting open polydiscs of the same radius are equal,

$$\mathfrak{o}(e, \gamma) = \mathfrak{o}(x, \gamma) \subseteq A_a.$$

Thus $D' \times \mathfrak{o}(e, \gamma)$ is an open subset of $X$ contained in $A$ and meeting $S$, a contradiction. Therefore $S_1$ is finite. Away from $S_1$, every point of $A$ is interior relative to $X$, which proves the open case and, by relative complementation, the closed case. □

**Resolved referee R045 [Gap; revised] – The fiberwise openness proof needs reconstruction.** *Original source line 1173.* The displayed proof contains genuine mistakes, but the conclusion has a repair. Retain $d = \dim_{\mathrm{VF}}(S) < n$ instead of replacing it by $n - 1$. Define

$$\alpha(a, x) = \min\{\beta \geq 0 : \mathfrak{o}(x, \beta) \subseteq A_a\} \quad ((a, x) \in S).$$

This minimum exists: for a nonempty complement it is the maximum of 0 and the supremum of the valuative distances from $x$ to that complement; for a full fiber it is 0. The original unrestricted minimum fails for full fibers. If $S_1$ is infinite, choose a smooth algebraic chart of a stratum of $S$ whose projection to the first coordinate is nonconstant. In characteristic zero, a separating coordinate projection, including the first coordinate, is generically étale; a local branch over an open polydisc is therefore definable and continuous. Pull $\alpha$ back to this chart and shrink to an open polydisc where it is constant, say $\epsilon$, using generic local constancy of maps into $\Gamma$. Fix the remaining chart coordinates and shrink the first-coordinate disc $D$ until the resulting section $a \mapsto (a, s(a)) \in S$ satisfies $s(a) \in \mathfrak{o}(s(a_0), \epsilon)$. Then

$$D \times \mathfrak{o}(s(a_0), \epsilon) \subseteq A,$$

contradicting $(a_0, s(a_0)) \in S$. This proves that $S_1$ is finite, including when $d < n - 1$. The contradiction concerns interior points of $A$, not an open polydisc inside $S$. For the closed case and the next proposition, formulate the lemma over a fixed definable base $I \subseteq \mathrm{VF}$, allowing empty fibers; passage between $I$ and the actual projection costs only its finite unary boundary.

**Author resolution.** The author approves the independently reconstructed bad-locus proof. A definable family of valid radii and the RV-indexed union principle supply one radius on a whole disc of base points. Finite nonempty choices and unary generic continuity give an open subset of $A$ meeting $S$, the required contradiction. No codimension-one assumption or minimum-radius choice is needed. The statement is retained, and working over a fixed unary base with empty fibers allowed justifies the closed case by relative complement.

**Unapplied R046 proof-detail proposal after independent main-text review.** The original implication is valid through a standard relative-closure step. The statement and the phrase "as in Lemma 2.54" are retained. The green proof takes ambient closures of fiber graphs, applies the fixed-base closed-family lemma in the two affine charts, and intersects back with the domain before invoking the closed-graph criterion. The target wording is made consistent with the already intended projective valuation topology. This is an expansion of a compressed argument, with no new hypothesis or claim of a false proposition.

**Proposition 2.55.** *Let $A$ be as in Lemma 2.54 and $f : A \longrightarrow \mathrm{VF}$ a definable function such that for each $a \in A_1$, the induced function $f_a : A_a \longrightarrow \mathrm{VF}$ is continuous. Then there is a finite definable set $A' \subseteq A_1$ such that $f$ is continuous on $\bigcup_{a \in A_1 \smallsetminus A'} a \times A_a$.*

For the closed-graph argument, the projective line $\mathbb{P}^1(\mathrm{VF})$ is understood with the valuation topology, as in Lemma 2.50.

*Proof.* Put $I = A_1$ and write $H_a$ for the graph of $f_a$. Let

$$G_a = \mathrm{Cl}_{\mathrm{VF}^{n-1}\times\mathbb{P}^1(\mathrm{VF})}(H_a), \qquad G = \bigcup_{a\in I}\{a\}\times G_a.$$

This family is definable: fiberwise closure is expressed uniformly by the valuative neighborhood formula in each affine chart. Every $G_a$ is closed in $\mathrm{VF}^{n-1}\times\mathbb{P}^1(\mathrm{VF})$.

Apply the fixed-base version proved in Lemma 2.54, with empty fibers allowed, to $G$ in each of the two standard affine charts of $\mathbb{P}^1(\mathrm{VF})$. Outside the union $I'$ of the resulting two finite definable exceptional sets, $G$ is closed in

$$(I\setminus I')\times \mathrm{VF}^{n-1}\times\mathbb{P}^1(\mathrm{VF}),$$

since closedness is local on this open chart cover. Fiberwise continuity gives

$$H_a = G_a \cap \big(A_a\times\mathbb{P}^1(\mathrm{VF})\big).$$

Consequently, the graph of the restricted function is closed relative to

$$\big(A\cap((I\setminus I')\times\mathrm{VF}^{n-1})\big)\times\mathbb{P}^1(\mathrm{VF}).$$

The closed-graph criterion (Lemma 2.50) proves the asserted continuity. This spells out the relative-closure step in the original implication. □

**Referee R046 [Gap; revised] – Relative graph closedness is insufficient for the cited lemma.** *Original source line 1188.* Continuity of $f_a$ makes its graph closed in $A_a\times\mathbb{P}^1(\mathrm{VF})$, whereas the preceding closed-fiber lemma concerns ambient closed fibers. Define instead

$$G_a = \overline{\mathrm{graph}(f_a)} \subseteq \mathrm{VF}^{n-1}\times\mathbb{P}^1(\mathrm{VF}).$$

Use the preceding lemma in its fixed-base form over $I = A_1$, allowing empty fibers, in each of the two affine charts of $\mathbb{P}^1$. Equivalently, also discard the finite unary boundaries of the chart projections when applying the lemma as currently stated. Outside the union of these finite exceptional sets, $G$ is relatively closed over $I$. Fiberwise continuity gives $\mathrm{graph}(f) = G\cap(A\times\mathbb{P}^1)$, so the graph is closed in $A\times\mathbb{P}^1$, and the compactified closed-graph criterion applies. This proves the proposition without an open/closed-fiber hypothesis on $A_a$; clarify the phrase 'as in Lemma 2.54' accordingly.

This is analogous to [4, Corollary 6.2.4] in the $o$-minimal environment and [22, Lemma 2.38] in the $T$-convex environment.

## 3. Hrushovski-Kazhdan style integral

**Definition 3.1.** The Grothendieck semigroup $\mathbf{K}_+\mathcal{C}$ of a category $\mathcal{C}$ endowed with a binary operation "$\smallsetminus$" and a binary relation "$\subseteq$" (subobject) is the free semigroup generated by the isomorphism classes of $\mathcal{C}$, subject to the usual scissor relation $[A\smallsetminus B]+[B]=[A]$, when $B\subseteq A$, where $[A]$, $[B]$ denote the isomorphism classes of the objects $A$, $B$, and "$\smallsetminus$" and "$\subseteq$" are compatible with isomorphisms between objects, usually they are just set subtraction and set inclusion. Sometimes $\mathcal{C}$ is also equipped with a binary operation (for example, cartesian product) that induces multiplication in $\mathbf{K}_+\mathcal{C}$, in which case $\mathbf{K}_+\mathcal{C}$ becomes a (commutative) semiring. The formal groupification of $\mathbf{K}_+\mathcal{C}$, which is then a ring, is denoted by $\mathbf{K}\mathcal{C}$.

**Referee R047 [Clarification; revised] – Commutative Grothendieck construction.** *Original source line 1199.* Optional precision: call this the free *commutative monoid* on the isomorphism classes, subject to the scissor relations, with the empty object representing 0. The final phrase "which is then a ring" can be read as referring to the preceding case where multiplication has been supplied; one